



REST API dengan Lumen (CRUD)

Database & Tabel

- Database: json
- Struktur tabel “wisata”

Nama	Jenis ?	Panjang/Nilai ?
id_wisata	INT ▼	2
kota	VARCHAR ▼	10
landmark	VARCHAR ▼	100
tarif	VARCHAR ▼	10

id_wisata = primary_key & Auto Increment

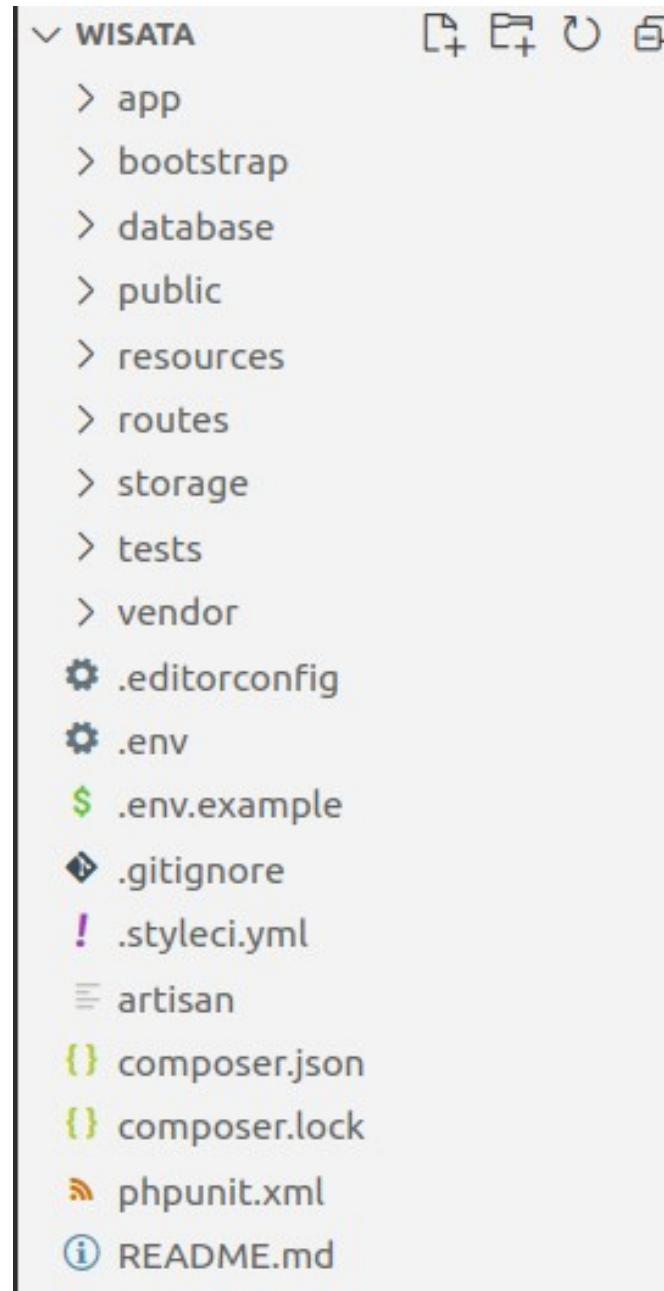
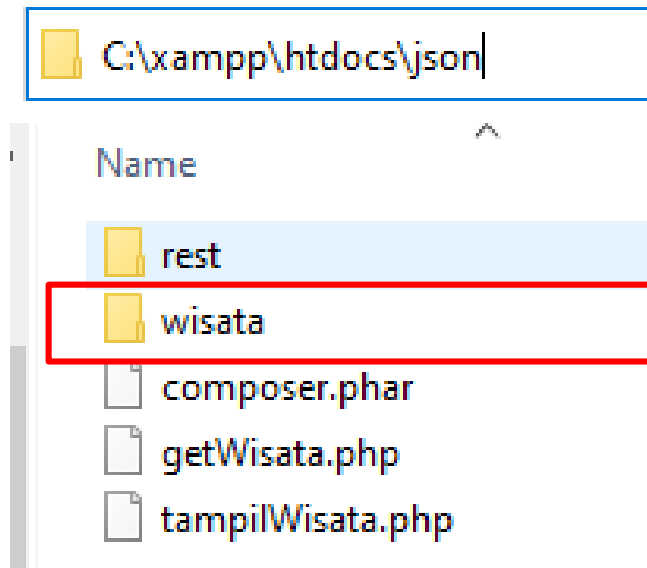
Isi tabel

KOTA	LANDMARK	TARIF
SEMARANG	LAWANG SEWU	20000
YOGYAKARTA	PRAMBANAN	35000
MAGELANG	BOROBUDUR	45000
SURAKARTA	PGS	GRATIS

Server Requirements

- PHP $\geq$ 7.3
- OpenSSL PHP Extension
- PDO PHP Extension
- Mbstring PHP Extension
- Composer
 - Download composer
-- atau --
 - Install composer

Folder Project hasil instalasi



Konfigurasi APP_URL dan Database (.env)

- File .env (wisata\.env)

⚙ .env

```
1 APP_NAME=Lumen
2 APP_ENV=local
3 APP_KEY=
4 APP_DEBUG=true
5 APP_URL=http://localhost:8000
6 APP_TIMEZONE=UTC
7
8 LOG_CHANNEL=stack
9 LOG_SLACK_WEBHOOK_URL=
```

```
11 DB_CONNECTION=mysql
12 DB_HOST=127.0.0.1
13 DB_PORT=3306
14 DB_DATABASE=json
15 DB_USERNAME=root
16 DB_PASSWORD=
17
18 CACHE_DRIVER=file
19 QUEUE_CONNECTION=sync
20
```

Aktivasi fungsi Eloquent dan Facades

- Edit file: wisata\bootstrap\app.php
- Uncomment (hilangkan //) baris berikut

```
bootstrap > app.php
25
26 // $app->withFacades();
27
28 // $app->withEloquent();
29
```

- menjadi

```
bootstrap > app.php
25
26 $app->withFacades();
27
28 $app->withEloquent();
```

Buat model “Wisata”

- Buat file: wisata\app\Models\Wisata.php

```
<?php
namespace App\Models;
use Illuminate\Database\Eloquent\Model;

class Wisata extends Model
{
    protected $table = 'wisata';
    protected $primaryKey = 'id_wisata';
    public $timestamps = false;

    /**
     * The attributes that are mass assignable.
     *
     * @var array
     */
    protected $fillable = [
        'kota', 'landmark', 'tarif'
    ];

    /**
     * The attributes excluded from the model's JSON form.
     *
     * @var array
     */
    protected $hidden = [];
}
```


Buat controller “WisataController”

- Buat file: wisata\app\Http\Controllers\WisataController.php

```
<?php
```

```
namespace App\Http\Controllers;
```

```
use Illuminate\Http\Request;
```

```
use App\Models\Wisata;
```

```
class WisataController extends Controller  
{
```

```
    public function showAll()  
    {
```

```
        return response()->json(Wisata::all());  
    }
```

```
    public function showOne($id)  
    {
```

```
        return response()->json(Wisata::find($id));  
    }
```

Buat controller “WisataController”

- Lanjutan....

```
public function create(Request $request)
{
    $Wisata = Wisata::create($request->all());

    return response()->json($Wisata, 201);
}

public function update($id, Request $request)
{
    $Wisata = Wisata::findOrFail($id);
    $Wisata->update($request->all());

    return response()->json($Wisata, 200);
}

public function delete($id)
{
    Wisata::findOrFail($id)->delete();
    return response('Deleted Successfully', 200);
}

}
```

Konfigurasi Routing

- Edit file: wisata\routes\web.php
- Tambahkan routing berikut

```
$router->group(['prefix' => 'api'], function () use ($router) {  
    $router->get('wisata', ['uses' => 'WisataController@showAll']);  
    $router->get('wisata/{id}', ['uses' => 'WisataController@showOne']);  
    $router->post('wisata', ['uses' => 'WisataController@create']);  
    $router->delete('wisata/{id}', ['uses' => 'WisataController@delete']);  
    $router->put('wisata/{id}', ['uses' => 'WisataController@update']);  
});
```

routes >  web.php

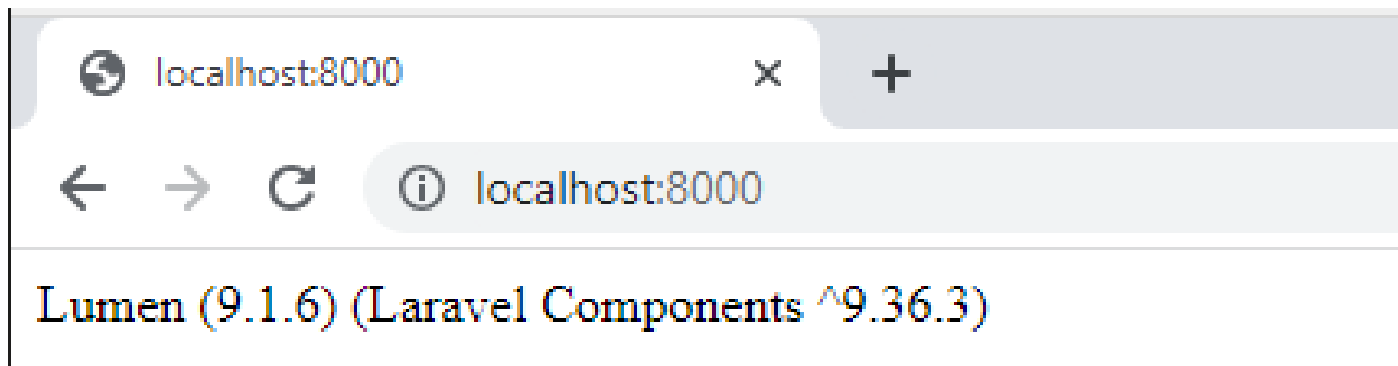
```
19  
20 $router->group(['prefix' => 'api'], function () use ($router) {  
21     $router->get('wisata', ['uses' => 'WisataController@showAll']);  
22     $router->get('wisata/{id}', ['uses' => 'WisataController@showOne']);  
23     $router->post('wisata', ['uses' => 'WisataController@create']);  
24     $router->delete('wisata/{id}', ['uses' => 'WisataController@delete']);  
25     $router->put('wisata/{id}', ['uses' => 'WisataController@update']);  
26 });
```

Ujicoba REST API Lumen menggunakan POSTMAN

- Jalankan server Database MySQL
- Jalankan server web dengan perintah berikut:
 - `php -S localhost:8000 -t public`

```
c:\xampp\htdocs\json\wisata>php -S localhost:8000 -t public  
[Fri Jun 16 21:18:07 2023] PHP 8.0.28 Development Server (http://localhost:8000) started
```

Akses alamat localhost:8000 melalui browser



Tampilan request GET: api/wisata

GET

http://localhost:8000/api/wisata

ParamsAuthorizationHeaders (6)BodyPre-request ScriptTestsSettings

BodyCookiesHeaders (7)Test Results

200 OK

PrettyRawPreviewVisualize

JSON

```
1 [
2   {
3     "id_wisata": 1,
4     "kota": "SEMARANG",
5     "landmark": "LAWANG SEWU",
6     "tarif": "200000000"
7   },
8   {
9     "id_wisata": 2,
10    "kota": "YOGYAKARTA",
11    "landmark": "PRAMBANAN",
12    "tarif": "35000"
13  },
14 ]
```

Tampilan request GET: api/wisata/{id}

GET



http://localhost:8000/api/wisata/1

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Body

Cookies

Headers (7)

Test Results



200 OK



Pretty

Raw

Preview

Visualize

JSON



```
1 {
2   "id_wisata": 1,
3   "kota": "SEMARANG",
4   "landmark": "LAWANG SEWU",
5   "tarif": "200000000"
6 }
```

Tampilan request POST: api/wisata/

POST

http://localhost:8000/api/wisata/

Params

Authorization

Headers (8)

Body ●

Pre-request Script

Tests

Settings

☐ none ☒ form-data ☐ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

☒	kota	semarang	
☒	landmark	pagoda	
☒	tarif	25000	

Body

Cookies

Headers (7)

Test Results



201 Created

Pretty

Raw

Preview

Visualize

JSON ▼



```
1  {
2    "kota": "semarang",
3    "landmark": "pagoda",
4    "tarif": "25000",
5    "id_wisata": 45
6  }
```

Tampilan request PUT: api/wisata/{id}

PUT

http://localhost:8000/api/wisata/45

Params

Authorization

Headers (8)

Body

Pre-request Script

Tests

Settings

☐ none ☐ form-data ☒ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

☒	kota	semarang	
☒	landmark	pagoda Avalokitesvara	
☒	tarif	35000	

Body

Cookies

Headers (7)

Test Results

 200 OK

Pretty

Raw

Preview

Visualize

JSON



```
1 {  
2   "id_wisata": 45,  
3   "kota": "semarang",  
4   "landmark": "pagoda Avalokitesvara",  
5   "tarif": "35000"  
6 }
```


Tampilan request DELETE: api/wisata/{id}

DELETE

▼

http://localhost:8000/api/wisata/45

Params

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

☒ none

☐ form-data

☐ x-www-form-urlencoded

☐ raw

☐ binary

☐ GraphQL

This request does not have a body

Body

Cookies

Headers (7)

Test Results

200 OK

Pretty

Raw

Preview

Visualize

HTML ▼

↺

1

Deleted Successfully